

Week-1

Discussion questions

- What is Naive Bayes Classification?
- Naive Bayes Classification - Example
- What is Gaussian Naive Bayes Classifier?
- Naive Bayes Classifier - Pros and Cons
- What is K Nearest Neighbours (KNN) and how does it work?
- Types of Distance metrics
- Features of KNN
- KNN - Pros and Cons

What is Naive Bayes Classification?

- Naïve Bayes classification is a form of classification that relies on the Bayes theorem.
- Bayes theorem is a theorem in probability that tells us how we would revisit the probability of an event given that we have more information.
- Bayes theorem takes information and constructs beliefs for the future.
- In simple terms, a Naive Bayes classifier assumes that the presence of a particular feature in a class is unrelated to the presence of any other feature.

What is a Naive Bayes Classification?

- It is called “naive” due to the assumption that the features in the dataset are mutually independent.
- In real world, the independence assumption is often violated, but naïve Bayes classifiers still tend to perform very well.
- Idea is to factor all available evidence in form of predictors into the naïve Bayes rule to obtain more accurate probability for class prediction.
- It estimates conditional probability which is the probability that something will happen, given that something else has already occurred. For e.g. the given mail is likely a spam given appearance of words such as “prize”

Naive Bayes Classification - Example

A -> event

- $P(A) = 0.5$
- $P(\bar{A}) = 1 - P(A)$

For eg: Rolling a 6 face dice

A -> event that you roll a 1

B -> event that you roll an odd number

- $P(A) = 1/6$
- $P(B) = 3/6 = 1/2$

Conditional Probability : Probability of event A conditioned on the probability that event B happened

$$P(A/B) = P(A \text{ and } B) / P(B) = (1/6) / (1/2) = 1/3$$

Naive Bayes Classification - Example

A -> event that you roll a 1 on dice 1

B -> event that you roll a 1 on dice 2

$$P(A \text{ and } B) = 1/36 = P(A) * P(B)$$

$$P(A/B) = 1/6 = P(A \text{ and } B) / P(B) = (P(A)*P(B)) / P(B) = P(A)$$

$$P(B/A) = 1/6 = P(B)$$

Bayes Theorem

Bayes theorem

- $P(A|B)$, reads “A given B,” represents the probability of A if B was known to have occurred.
- In many situations we would like to understand the relation between $P(A|B)$ and $P(B|A)$.
- You are planning an outdoor event tomorrow. When it actually rains, the weatherman correctly forecasts rain 90% of the time. When it doesn't rain, he incorrectly forecasts rain 10% of the time. Historically it has rained only 5 days each year. Unfortunately, the weatherman has predicted rain for tomorrow. What is the probability that it will rain tomorrow?

$$P(A|B) = \frac{P(B|A) P(A)}{P(B)}$$

Naive Bayes Classification - Example

- Will my flight be on time? It is sunny, hot, Normal Humidity, and not windy!
- Data from the last several times we took this flight.

OUTLOOK	TEMPERATURE	HUMIDITY	WINDY	Flight On Time
Rainy	Hot	High	0	No
Rainy	Hot	High	1	Yes
Overcast	Hot	High	0	Yes
Sunny	Mild	High	0	No
Sunny	Cool	Normal	0	Yes
Sunny	Cool	Normal	1	No
Overcast	Cool	Normal	1	Yes
Rainy	Mild	High	0	No
Rainy	Cool	Normal	0	Yes
Sunny	Mild	Normal	0	Yes
Rainy	Mild	Normal	1	Yes
Overcast	Mild	High	1	Yes
Overcast	Hot	Normal	0	Yes
Sunny	Mild	High	1	No

Naive Bayes Classification - Example

1. $P(\text{Flight on time} \mid \text{Sunny, Hot, Normal humidity, 0}) =$
2. $P(\text{Yes} \mid \text{S, H, N, 0}) = (P(\text{Sunny} \mid \text{Yes}) * P(\text{Hot} \mid \text{Yes}) * P(\text{Normal} \mid \text{Yes}) * P(0 \mid \text{Yes}) * P(\text{Yes})) / (P(\text{Sunny}) * P(\text{Hot}) * P(\text{Normal}) * P(0))$
3. $P(\text{Yes}) = 9/14$
4. $P(\text{Yes} \mid \text{S, H, N, 0}) = (P(\text{Sunny} \mid \text{Yes}) * P(\text{Hot} \mid \text{Yes}) * P(\text{Normal} \mid \text{Yes}) * P(0 \mid \text{Yes}) * P(\text{Yes})) / ((P(\text{Sunny}) * P(\text{Hot}) * P(\text{Normal}) * P(0))$

$$= ((2/9 * 3/9 * 6/9 * 5/9) * (9/14)) / ((5/14) * (4/14) * (7/14) * (8/14))$$
5. $P(\text{S, H, N, 0} \mid \text{Yes}) = 0.604938$
6. $P(\text{Flight on time} \mid \text{Sunny, Hot, Normal humidity, 0}) = 60\%$

What is Gaussian Naive Bayes Classifier?

- When some of our independent variables are continuous we cannot calculate conditional probabilities!
- In Gaussian Naive Bayes, continuous values associated with each feature (or independent variable) are assumed to be distributed according to a Gaussian distribution.
- All we would have to do is estimate the mean and standard deviation of the continuous variable.

Naive Bayes Classifier - Pros and Cons

Pros

- Simple, Fast in processing and effective
- Does well with noisy data and missing data
- Requires few examples for training (assuming the data set is a true representative of the population)
- Easy to obtain estimated probability for a prediction

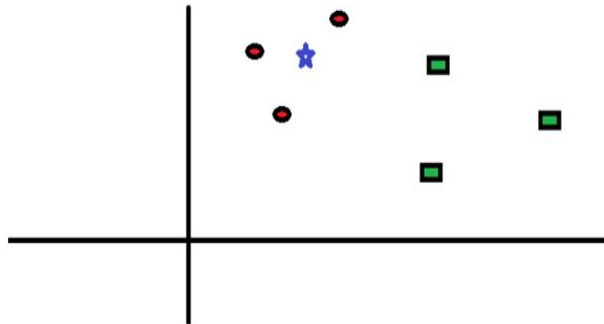
Cons

- Relies on and often incorrect assumption of independent features
- Not ideal for data sets with large number of numerical attributes
- Estimated probabilities are less reliable in practice than predicted classes
- If rare events are not captured in the training set but appears in the test set the probability calculation will be incorrect

What is KNN and how does it work?

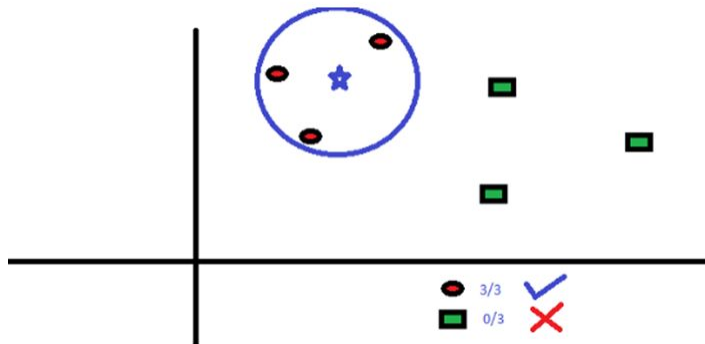
- KNN (K Nearest Neighbors) is one of many (supervised learning) algorithms used in data mining and machine learning.
- It's a classifier algorithm where the learning is based “how similar” is a data (a vector) from other.
- Let's take a simple case to understand this algorithm. Following is a spread of red circles (RC) and green squares (GS) :

What is KNN and how does it work?



You intend to find out the class of the blue star (BS). BS can either be RC or GS and nothing else. The “K” in KNN algorithm is the nearest neighbors we wish to take vote from. Let’s say $K = 3$. Hence, we will now make a circle with BS as center just as big as to enclose only three data points on the plane. Refer to following diagram for more details:

What is KNN and how does it work?



The three closest points to BS is all RC. Hence, with good confidence level we can say that the BS should belong to the class RC. Here, the choice became very obvious as all three votes from the closest neighbor went to RC. The choice of the parameter K is very crucial in this algorithm.

Classification Steps

1. Receive an unclassified data
2. Measure the distance (Euclidian, Manhattan, Minkowski or Weighted) from the new data to all others data that is already classified
3. Gets the K(K is a parameter that you define) smaller distances
4. Check the list of classes had the shortest distance and count the amount of each class that appears
5. Takes as correct class the class that appeared the most times
6. Classifies the new data with the class that you took in step 5

Types of Distance metrics

Distance measure is important

- Most commonly distance is measured using Euclidean distance
- We should always Normalize data
- Other distance measurement methods include
 - Manhattan distance
 - Minkowski distance
 - Mahalanobis distance
 - Cosine similarity

Types of Distance metrics

To calculate the distance between two points is very simple, there are several ways to get this value, here we will use the Euclidean distance.

The Euclidean distance formula is:

$$\begin{aligned}d(\mathbf{p}, \mathbf{q}) &= d(\mathbf{q}, \mathbf{p}) = \sqrt{(q_1 - p_1)^2 + (q_2 - p_2)^2 + \cdots + (q_n - p_n)^2} \\&= \sqrt{\sum_{i=1}^n (q_i - p_i)^2}.\end{aligned}$$

Types of Distance metrics

- **Manhattan distance:** The distance between two points measured along axes at right angles. In a plane with p_1 at (x_1, y_1) and p_2 at (x_2, y_2) , it is $|x_1 - x_2| + |y_1 - y_2|$.
- **Minkowski distance:** It is a metric in a normed vector space. Minkowski distance is used for distance similarity of vector. Given two or more vectors, find distance similarity of these vectors. Mainly Minkowski distance is applied in machine learning to find out distance similarity.
- **Cosine similarity:** It is a measure of similarity between two non-zero vectors of an inner product space that measures the cosine of the angle between them. The cosine of 0° is 1, and it is less than 1 for any angle in the interval $(0, \pi]$ radians.

Variants of KNN

Radius Neighbor Classifier

- Implements learning based on number of neighbors within a fixed radius r of each training point, where r is a floating point value specified by the user
- May be a better choice when the sampling is not uniform. However, when there are many attributes and data is sparse, this method becomes ineffective due to curse of dimensionality.

KD Tree nearest neighbour

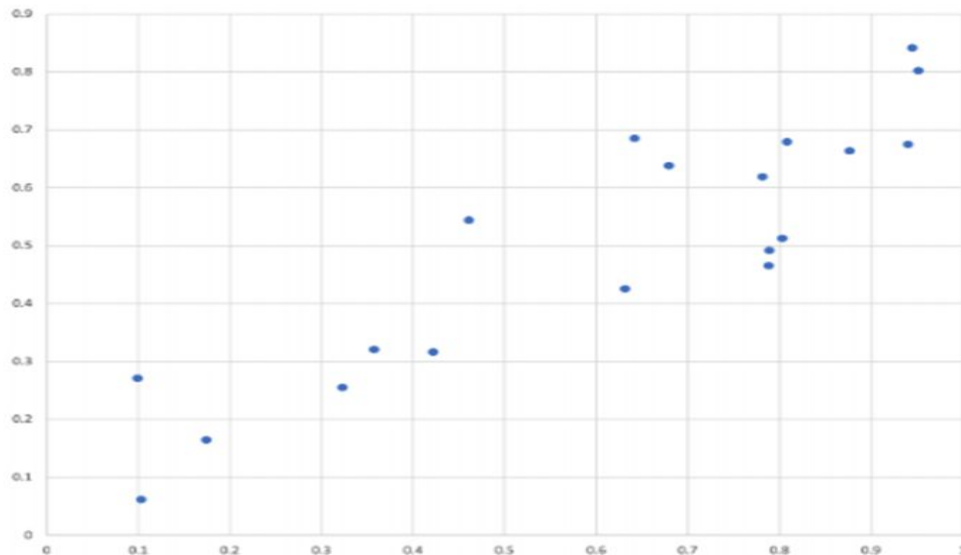
- Approach helps reduce the computation time.
- Very effective when we have large data points but still not too many dimensions

Features of KNN

- It does not construct a “model”. Known as a non-parametric method.
- Classification is computed from a simple majority vote of the nearest neighbors of each point
- Suited for classification where relationship between features and target classes is numerous, complex and difficult to understand and yet items in a class tend to be fairly homogenous on the values of attributes
- Not suitable if the data is too noisy and the target classes do not have clear demarcation in terms of attribute values
- Can also be used for regression.

KNN for regression

The Neighbors based algorithm can also be used for regression where the labels are continuous data and the label of query point can be average of the labels of the neighbors.



K Nearest Neighbours - Pros and Cons

Pros

- Makes no assumptions about distributions of classes in feature space
- Can work for multi classes simultaneously
- Easy to implement and understand
- Not impacted by outliers.

Cons

- Fixing the optimal value of K is a challenge
- Will not be effective when the class distributions overlap
- Does not output any models. Calculates distances for every new point(lazy learner)
- Computationally intensive



Happy Learning !

